Sri Lanka Institute Of Information Technology



IT3091 – MACHINE LEARNING

FINAL PROJECT REPORT

Healthcare – Early Readmission Risk Prediction

Track: Guided Data Track

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1. Executive Summary

This project investigates whether historical hospital encounter data can support the identification of encounters associated with readmission within 30 days. It follows the IT3091 Guided Data Track and uses the Diabetes 130 - US Hospitals for Years 1999 - 2008 dataset. The primary business lens is readmission risk; the intended stakeholders are hospital management, healthcare planning teams and hospital analytics teams.

The original dataset contains 101,766 encounters and 50 source variables. The notebook removed 2,426 records associated with specified death/hospice discharge codes or invalid gender, leaving 99,340 encounters. The target was defined as early_readmission = 1 when readmitted is “<30”, and 0 for “NO” or “>30”. The resulting target contains 11,314 early-readmission encounters (11.39%) and 88,026 non-early-readmission encounters (88.61%).

The workflow includes data-quality analysis, exploratory analysis, feature engineering, leakage review, patient-level train/test splitting, preprocessing pipelines, a majority-class baseline, and four candidate classifiers. A GroupShuffleSplit created 79,567 training encounters and 19,773 test encounters with zero patient overlap. Three-fold StratifiedGroupKFold was used for group-aware cross-validation.

Extra Trees was selected because it achieved the highest cross-validation PR-AUC (0.2141) and cross-validation F1 (0.2751) among the compared models. On the held-out test set at the selected threshold of 0.50, it achieved PR-AUC 0.2300, ROC-AUC 0.6783, precision 0.2061, recall 0.4581 and F1 0.2843. These results show modest ranking ability and a meaningful recall/false-positive trade-off, not clinical readiness. The model should only support human-reviewed prioritisation for discharge planning or follow-up.

2. Business Problem Framing and Task Formulation

2.1 Business problem

Hospitals need to plan discharge support and follow-up under limited staffing and resources. Historical encounters may contain signals associated with early readmission. The project tests whether these records can be used to rank or flag encounters for additional review, rather than automatically determine treatment or discharge decisions.

2.2 Problem-framing canvas

Element	**Project formulation**
Domain / track	Healthcare; Guided Data Track
Primary decision lens	Readmission risk
Stakeholders	Hospital management, healthcare planning, hospital analytics teams; clinical staff review any action
Decision need	Which encounters may warrant additional discharge planning or follow-up consideration?
Unit of analysis	Hospital encounter; multiple encounters can belong to one patient
Learning type / task	Supervised binary classification
Target	early_readmission: 1 = readmitted within 30 days (“<30”); 0 = “NO” or “>30”
Output	Predicted class and model probability/risk score
Intended use	Prioritise human review; not diagnosis or autonomous clinical decision-making
Primary selection lens	Cross-validation PR-AUC for the imbalanced positive class; F1 and recall/precision trade-offs also reviewed
Why this lens	The positive class is only 11.39%; PR-AUC is more informative than accuracy alone for positive-class retrieval.

2.3 Scope and assumptions

- The model estimates association with the recorded early-readmission label, not whether a readmission was preventable.
- The data are historical US hospital records from 1999 - 2008; performance may not transfer to present-day hospitals or other countries.
- Predictions are only meaningful when predictors are available at the intended decision point. The notebook states that retained features are available by discharge; an implementation must verify this against real operational data.

3. Workflow Diagram and Decision Log

3.1 End-to-end workflow

Stage	Activity	Output / control
1. Frame	Choose readmission-risk lens and binary target	Defined stakeholder decision and unit of analysis
2. Understand	Inspect rows, columns, types, missingness, duplicates and repeat patients	Data-quality summary and dictionary
3. Prepare target	Remove specified death/hospice and invalid-gender rows; map target	99,340 rows; 11.39% positive
4. Explore	Examine target, age, length of stay, prior visits and gender	EDA insight log; association not causation
5. Engineer	Create prior-visit totals/flag, diagnosis groups and numeric age	More interpretable, task-relevant features
6. Split	Group by patient; hold out 20%	Zero patient overlap
7. Transform	Numeric scaling; categorical encoding; missing-value semantics	Pipeline fitted only within model workflow
8. Compare	Dummy, Logistic Regression, Decision Tree, Random Forest, Extra Trees	Consistent test and group-aware CV comparisons
9. Evaluate	PR-AUC, ROC-AUC, precision, recall, F1 and confusion matrix	Critical assessment beyond accuracy
10. Recommend	Select Extra Trees; propose human-reviewed prioritisation	Limitations and validation requirements

3.2 Key decisions

The project's main decisions are consolidated in Appendix A. The most consequential are: defining the target before modelling; excluding identifiers and the original outcome from predictors; grouping encounters by patient during splitting and cross-validation; handling missingness according to its meaning; using class weighting rather than synthetic resampling; and selecting the final model using metrics suitable for an imbalanced target.

4. Dataset, Data Dictionary and Data Understanding

4.1 Dataset overview

Property	Notebook - supported value
Dataset	Diabetes 130-US Hospitals for Years 1999 - 2008
Main file	diabetic_data.csv
Mapping file	IDS_mapping.csv
Original size	101,766 rows × 50 source columns
After row exclusions	99,340 rows; 2,426 removed
Unique patients after exclusions	69,987
Exact duplicate rows	0
Predictor matrix before one-hot encoding	49 columns; 11 numerical and 38 categorical
Target prevalence after exclusions	11.39% early readmission

Each row represents a hospital encounter. Variables cover demographics, admission and discharge codes, diagnoses, procedures, medication-related information, prior healthcare utilisation and the readmission outcome. IDS_mapping.csv provides mappings for coded admission, discharge and admission-source identifiers.

4.2 Target definition

Original readmitted value	Mapped target	Interpretation
<30	1	Early readmission within 30 days
NO	0	No recorded readmission
>30	0	Readmission after 30 days; not part of the early class

Target counts are 11,314 positive and 88,026 negative encounters. This differs from the original report's target count because the final notebook removes 2,426 rows before target modelling. The revised report uses the notebook figures.

4.3 Data dictionary

Appendix B provides a compact variable-level dictionary for the source columns, with definitions and modelling treatment. The final notebook also generates a data_dictionary.csv containing observed type, missing count, missing percentage and unique-value count. Coded IDs are treated as categorical rather than continuous measurements.

5. Data Quality Assessment and EDA

5.1 Missingness and duplicates

Variable(s)	Missing count / rate in original data	Decision and reasoning
weight	98,569 / 96.86%	Dropped because almost all values are missing; imputation would be weakly supported.
max_glu_serum	96,420 / 94.75%	Missing represented as “Not tested” under the notebook’s semantic assumption that absence can reflect no test ordered.
A1Cresult	84,748 / 83.28%	Missing represented as “Not tested” under the same assumption.
medical_specialty	49,949 / 49.08%	Missing represented as “Unknown”; missingness may reflect unrecorded specialty.
payer_code	40,256 / 39.56%	Missing represented as “Unknown”.
race	2,273 / 2.23%	Missing represented as “Unknown”.
diag_3 / diag_2 / diag_1	1,423 / 1.40%; 358 / 0.35%; 21 / 0.02%	Diagnosis values were grouped into ICD-9 chapters; residual handling follows notebook pipeline.
Exact duplicates	0	No exact duplicate rows were removed.

5.2 Repeated patients and leakage risk

The original data contain 71,518 unique patients among 101,766 encounters, and 16,773 patients have multiple encounters. After the notebook’s exclusions, 69,987 unique patients remain. A random row split could put encounters from one patient in both training and test sets, creating an optimistic evaluation. The notebook therefore uses patient_nbr only as a grouping variable and confirms zero patient overlap after the split.

5.3 EDA insight log

EDA observation from notebook	Interpretation / decision
Early readmission is 11.39% of retained encounters	Minority class; accuracy alone can hide poor positive-class detection. Report PR-AUC, recall, precision and F1.
Rate rises with prior inpatient visits: 8.6% with 0 visits to 31.4% with 4+	Engineered total_prior_visits and has_prior_inpatient; association is not causal evidence.
Rate rises with prior emergency visits: 10.7% with 0 to 25.3% with 3+	Prior utilisation may carry predictive signal.
Rate rises with length of stay: 8.4% at 1 day to about 14 - 15% at 8 - 10 days	Keep time_in_hospital as a candidate predictor; do not infer that length of stay causes readmission.
Gender rates: Female 11.46%, Male 11.30%	Small descriptive difference; retain as a candidate feature but do not overstate importance.

6. Preprocessing and Feature Engineering

6.1 Row and target handling

- Removed discharge disposition codes 11, 13, 14, 19, 20 and 21, interpreted in the notebook as death/hospice outcomes where readmission is not possible; removed three "Unknown/Invalid" gender rows. Total removed: 2,426.
- Constructed early_readmission from readmitted: "<30" → 1; "NO" and ">30" → 0.
- Excluded encounter_id, patient_nbr, readmitted and early_readmission from predictor features. patient_nbr was retained separately only for grouped splitting/validation.

6.2 Feature engineering

Feature / transformation	**Purpose**	**Caution**
total_prior_visits = inpatient + emergency + outpatient counts	Summarises prior utilisation into one measure	Correlated with component counts; evaluate for redundancy if further tuning.
Has_prior_inpatient = 1 if number_inpatient > 0 else 0	Captures whether there was any prior inpatient encounter	May overlap with number_inpatient; feature importance is associative.
diag_1_group, diag_2_group, diag_3_group	Maps raw diagnosis codes into ICD-9 chapter groups to reduce sparsity	Validate mapping rules and "other/unknown" treatment in code.
age_numeric	Transforms age-band text into a numeric representation for analysis/model input	Underlying age is grouped/banded, not exact age; numeric mapping assumes an order/approximation.

6.3 Missing values and encoding

- The “weight” field was dropped because 96.86% was missing.
- Missing race, payer_code and medical_specialty values were replaced with “Unknown”.
- Missing A1Cresult and max_glu_serum values were replaced with “Not tested”, based on the notebook’s interpretation that missingness can indicate a test was not ordered. This is an assumption and should be checked with domain experts.
- Categorical features use one-hot encoding with min_frequency = 0.01, grouping rare levels to control dimensionality and handle_unknown="ignore" to avoid failures for unseen categories.
- Numerical features are standardised. Scaling is particularly useful for Logistic Regression; tree-based models are generally less scale-sensitive.
- Class weighting (“balanced” / “balanced_subsample”, depending on estimator) increases the influence of the minority class without generating synthetic patient records.

6.4 Leakage controls and pipeline

Preprocessing is implemented within a scikit-learn ColumnTransformer/Pipeline, so learned transformations are fitted within the model workflow rather than on the full dataset before splitting. The original readmitted outcome and target are excluded from predictors. The patient identifier is not a feature. These controls reduce leakage risk, though operational deployment still requires a time-of-prediction audit of every retained field.

7. Model Strategy and Comparison

7.1 Candidate models

Method	Role / rationale	Main limitation
Dummy Classifier	Majority-class reference baseline	Accuracy is high but it predicts no early readmissions.
Logistic Regression	Interpretable linear baseline; scaled inputs are useful	May not capture complex nonlinear relationships without feature expansion.
Decision Tree	Nonlinear rules and potentially interpretable splits	Can overfit; here it has lower PR-AUC than the selected model.
Random Forest	Bagged tree ensemble; nonlinear patterns and interactions	Less transparent than a single tree; positive-class recall at default threshold is lower here.
Extra Trees	Randomised tree ensemble; strong tabular-data alternative	Less directly interpretable; feature importance is not causal.

7.2 Held - out test comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Extra Trees	0.7382	0.2061	0.4581	0.2843	0.6783	0.2300
Logistic Regression	0.6606	0.1821	0.5700	0.2760	0.6688	0.2208
Random Forest	0.8115	0.2425	0.3115	0.2727	0.6770	0.2292
Decision Tree	0.6126	0.1633	0.5856	0.2554	0.6398	0.2062
Dummy Baseline	0.8865	0.0000	0.0000	0.0000	N/A	N/A

The Dummy Classifier has the highest accuracy because it predicts the majority class; it has zero recall and F1 for early readmission. This demonstrates why accuracy is not a defensible selection criterion on its own.

7.3 Group-aware cross-validation comparison

Model	CV Accuracy	CV Precision	CV Recall	CV F1	CV ROC-AUC	CV PR-AUC
Random Forest	0.8193	0.2364	0.2618	0.2482	0.6661	0.2112
Extra Trees	0.7463	0.2040	0.4223	0.2751	0.6630	0.2141
Logistic Regression	0.6563	0.1783	0.5584	0.2703	0.6597	0.2081
Decision Tree	0.6175	0.1649	0.5725	0.2550	0.6322	0.1922

7.4 Selection rationale

Extra Trees was selected because it has the highest cross-validation PR-AUC (0.2141) and highest cross-validation F1 (0.2751) among the candidate models. PR-AUC was prioritised because the positive class is uncommon and precision - recall performance directly reflects the positive-class retrieval problem. Random Forest has a slightly higher CV ROC-AUC (0.6661 versus 0.6630) and higher test precision (0.2425 versus 0.2061), while Extra Trees has higher test recall and F1. The models are close; the chosen model is best under the stated selection criterion, not best on every metric.

8. Evaluation, Validation and Critical Judgement

8.1 Split and validation

Item	**Final notebook result**
Split method	GroupShuffleSplit; 20% test; random_state=42
Training encounters / patients	79,567 encounters / 55,989 patients
Testing encounters / patients	19,773 encounters / 13,998 patients
Patient overlap	0
Cross-validation	3-fold StratifiedGroupKFold with patient groups; shuffle=True; random_state=42
Threshold selection	Out-of-fold training predictions; F1 evaluated over thresholds; selected threshold 0.50
Test-set threshold use	The test set was not used to choose the final threshold; threshold table is diagnostic sensitivity analysis only.

8.2 Final model performance at threshold 0.50

Metric	Value	Interpretation
Precision	0.2061	About 20.6% of predicted early-readmission encounters were actual early readmissions at this threshold.
Recall	0.4581	About 45.8% of actual early-readmission encounters were identified.
F1-score	0.2843	Balances precision and recall; still modest.
ROC-AUC	0.6783	Modest ability to rank positive cases above negative cases across thresholds.
PR-AUC	0.2300	Above the positive-class prevalence of 0.1139, but still indicates substantial false positives.
Accuracy	0.7382	Not sufficient alone; depends strongly on class prevalence and threshold.

8.3 Confusion matrix

Actual class	Predicted not early	Predicted early
Not early readmission	13,569	3,960
Early readmission	1,216	1,028

At the selected threshold, the model correctly identifies 1,028 early-readmission encounters and misses 1,216. It also produces 3,960 false positives. In practice, false negatives may mean that some encounters needing support are not flagged, while false positives may increase review workload. The preferred operating point depends on staffing capacity and the relative cost of these errors.

8.4 Threshold trade-off

Threshold	Precision	Recall	F1	Practical implication
0.40	0.1577	0.7482	0.2605	More early cases identified, with many more false positives.
0.45	0.1793	0.6123	0.2774	Higher recall than 0.50; lower precision.
0.50	0.2061	0.4581	0.2843	Selected using training out-of-fold F1; test F1 is highest among shown thresholds.
0.55	0.2493	0.2972	0.2712	Fewer false positives, but substantially more missed early cases.

The notebook selected 0.50 based on training out-of-fold F1. The test sensitivity table is not used to retune the threshold. Although the test F1 is highest around 0.50 among displayed thresholds, the threshold should ultimately be chosen with stakeholders using explicit workload and harm trade-offs and then validated prospectively.

8.5 Critical judgement

- Performance is modest: PR-AUC 0.230 is better than the 0.1139 prevalence baseline, but precision is only 0.206 at threshold 0.50.
- The majority baseline's accuracy is misleading; a model should be judged against the intended decision, not accuracy alone.
- Cross-validation and held-out test metrics are reasonably close in broad ranking terms, but one split and three folds do not establish robust generalisation.
- Feature importance describes model reliance/association, not causal effect. The notebook itself cautions that impurity-based importance can favour binary features; gender's appearance in a feature ranking should not override the nearly equal descriptive rates.
- A probability threshold is an operational choice, not a universal clinical cutoff. Calibration, decision-curve/utility analysis and prospective validation are not established by the current notebook.

9. Recommendation and Stakeholder Value

9.1 Recommendation

Treat Extra Trees as a prototype decision-support model for prioritising encounters for human review. If tested in a real workflow, predictions could be considered alongside clinical judgement to identify encounters for enhanced discharge planning, medication reconciliation, care coordination or follow-up review. These are candidate workflow uses, not interventions proven by this project.

9.2 Proposed implementation pathway

1. Validate the model on recent data from the intended hospital and confirm that each feature is available at the intended prediction time.
2. Agree on the decision objective and review capacity with clinical and operational stakeholders; choose a threshold using false-negative and false-positive costs.
3. Audit performance, calibration and error rates across relevant demographic and clinical subgroups.
4. Run a prospective, human-supervised pilot before any operational use. Record overrides and outcomes.
5. Monitor data drift, class prevalence, precision/recall, workload and downstream outcomes; retrain only through a documented governance process.

9.3 Stakeholder value and risks

Stakeholder	Potential value	Risk / control
Hospital management	More structured planning of review capacity and discharge support	Avoid interpreting scores as a guaranteed prediction; monitor workload and utility.
Healthcare planning team	Consistent prioritisation signal for follow-up review	False positives can overload services; threshold should reflect capacity.
Clinical staff	Additional information to support, not replace, judgement	Human review, explanation of limitations, and no automatic treatment decisions.
Patients	Potentially more timely attention when support needs are identified	Privacy, fairness and transparent communication are required.

10. Responsible AI, Ethics and Limitations

10.1 Responsible AI

- Privacy and security: restrict access to healthcare data, use it only for authorised purposes and follow applicable data-governance rules.
- Fairness: evaluate subgroup performance and calibration before use; aggregate performance can hide poor results for specific groups.
- Human oversight: predictions must not independently determine discharge, treatment or access to care.
- Transparency: explain that the output is a historical-data-based risk estimate, not a diagnosis or causal explanation.
- Error awareness: false negatives may miss encounters for extra support; false positives may consume staff time or create undue concern.

10.2 Limitations

- Historical source data from 130 US hospitals covering 1999–2008 may not represent current clinical practice or other health systems.
- The binary target groups “NO” and “>30” together, so it answers a narrower question about readmission within 30 days rather than all readmission outcomes.
- Removing death/hospice discharge codes is logically motivated for readmission prediction but changes the population and should be transparently documented.
- Class imbalance contributes to low precision and modest F1; many positive predictions are false alarms.
- EDA patterns are observational and do not establish causal factors or preventability.
- No clinical validation, prospective trial, fairness audit, calibration assessment or operational utility analysis is demonstrated in the supplied notebook.
- The age field is a banded category transformed into a numeric representation; this can introduce approximation.
- The “Not tested” treatment of missing laboratory results assumes missingness reflects testing decisions; that assumption requires domain validation.

11. Reproducibility, Documentation and AI-Use Transparency

11.1 Reproducibility record

Item	Record / required practice
Notebook	Healthcare_Readmission_Final_ipynb (7).ipynb; contains code, preprocessing, modelling, CV, evaluation and final recommendation. Rename to a clear final submission filename before submission if desired.
Dataset	diabetic_data.csv; include the actual file in the submission package if permitted by the course.
Mapping file	IDS_mapping.csv; include to interpret coded identifiers.
Randomness controls	Notebook uses random_state=42 for split and model settings where specified.
Environment	Record Python, pandas, NumPy, scikit-learn and plotting-library versions in the final notebook before submission. The supplied report does not verify exact package versions.
Generated outputs	Notebook writes final_model_comparison.csv, cross_validation_results.csv and data_dictionary.csv to /content/. Confirm these files are saved and included if required.
Re-run check	Restart kernel/runtime and run all notebook cells in order; confirm outputs reproduce the report tables.

11.2 Required companion artefacts

Artefact	Purpose / status
Final notebook/code (.ipynb)	Executable analysis. Include as a separate submission file; it is not replaced by narrative code snippets in this report.
Dataset and IDS mapping	Source data and coded-variable reference, subject to course submission rules.
Data dictionary (.csv or .xlsx)	Notebook creates data_dictionary.csv with type, missingness and cardinality. Appendix B summarises definitions and decisions.
Decision log	Appendix A in this report; may also be submitted as a separate spreadsheet.
Model comparison	Tables in Section 7; notebook also saves final_model_comparison.csv and cross_validation_results.csv.
Three-minute YouTube demo	Required by assignment descriptor; not embedded in this report.
Individual A4 Personal Learning Journey	Separate individual submission required by descriptor.

11.3 AI-use transparency

Generative AI tools were used to support learning, code assistance/debugging, explanations of machine-learning concepts and improvement of report documentation. The group should verify every generated suggestion against the notebook and course materials, and remains responsible for the final code, analysis, interpretation and conclusions. This declaration should be edited if it does not accurately describe the group's actual use. Any institution-specific AI citation or disclosure rules take precedence.

12. Conclusion

The project formulated early readmission as a binary supervised classification problem and followed a coherent workflow from business framing through data understanding, feature engineering, patient-level validation, model comparison and critical evaluation. The target is imbalanced, making accuracy alone unsuitable. Extra Trees was selected under the stated priority of group-aware CV PR-AUC and F1, achieving test PR-AUC 0.2300, ROC-AUC 0.6783 and F1 0.2843 at threshold 0.50. The result supports further investigation of human-reviewed prioritisation, but does not establish clinical readiness. Local validation, subgroup assessment, calibration, threshold governance and prospective evaluation are required before real-world use.

References

1. Sri Lanka Institute of Information Technology. (2026). IT3091 Machine Learning - Group Assignment Descriptor, Version 1.
2. Diabetes 130-US Hospitals for Years 1999- 2008 dataset, supplied for the IT3091 Guided Data Track.
3. scikit-learn documentation for pipelines, ColumnTransformer, classification estimators, GroupShuffleSplit, StratifiedGroupKFold and classification metrics. Consult the version used by the final notebook and add the exact version/date to the submitted notebook.

Appendix A. Consolidated Decision Log

ID	Stage	Decision	Reason / evidence	Final choice
D01	Problem framing	Use readmission risk as primary lens	Matches the supplied healthcare scenario and available outcome	Readmission risk
D02	Target	Map “<30” to 1; “NO” and “>30” to 0	Focused binary task for early readmission	Binary early_readmission
D03	Population	Remove specified death/hospice discharge codes and invalid gender rows	Notebook assumes readmission cannot occur for death/hospice outcomes; changes population	2,426 rows removed
D04	Identifiers	Exclude encounter_id and patient_nbr from predictors	IDs are not appropriate predictive features; patient_nbr only groups rows	Use patient_nbr for grouping only
D05	Leakage	Exclude readmitted and early_readmission from X	Original outcome creates target; target itself must not be a predictor	No outcome leakage in feature matrix
D06	Missing marker	Replace “?” with missing	Placeholder is not a true category	NaN before semantic treatment
D07	Sparse feature	Drop weight (96.86% missing)	Too little observed information for reliable imputation	Drop weight
D08	Categorical missingness	Use “Unknown” for race, payer_code, medical_specialty	Preserves unrecorded status as a category	Explicit Unknown category
D09	Lab missingness	Use “Not tested” for A1Cresult and max_glu_serum	Reflects notebook assumption that missing can indicate no test ordered	Explicit Not tested category; validate assumption
D10	Codes	Treat admission/discharge/source IDs as categorical	Numeric codes do not represent quantitative magnitude	Categorical encoding
D11	Diagnosis coding	Group diag_1/2/3 into ICD-9 chapters	Reduces sparse raw code levels	diag_*_group features
D12	Feature engineering	Add total_prior_visits and has_prior_inpatient; derive age_numeric	Summarises utilisation and supports model input	Engineered features retained
D13	Encoding	One-hot encode with min_frequency=0.01 and ignore unknowns	Avoids artificial order and controls rare levels	Categorical pipeline
D14	Scaling	Standardise numeric predictors	Supports scale-sensitive Logistic Regression	StandardScaler
D15	Imbalance	Use class weights	Improves minority-class influence without synthetic data	Balanced weights / balanced_subsample
D16	Split	GroupShuffleSplit with	Avoids same patient in train	79,567 train; 19,773

		20% test and seed 42	and test	test; zero overlap
D17	Validation	3-fold StratifiedGroupKFold grouped by patient	Group-aware estimate while approximately preserving target proportions	Three folds
D18	Metric	Prioritise PR-AUC; consider F1 and precision/recall	Positive class is uncommon; accuracy alone is misleading	PR-AUC primary selection lens
D19	Threshold	Select threshold using training out-of-fold F1	Avoids choosing threshold on held-out test data	0.50
D20	Model selection	Select Extra Trees	Highest CV PR-AUC 0.2141 and CV F1 0.2751; RF slightly higher CV ROC-AUC	Extra Trees
D21	Deployment	Human-reviewed prioritisation only	Modest precision and no clinical validation	No autonomous decisions

Appendix B. Data Dictionary

The table summarises source-variable meaning and treatment in the supplied notebook. “Original source type” refers to the general conceptual type; the notebook’s observed pandas types may differ before transformations. Use IDS_mapping.csv for detailed coded-value labels and the generated data_dictionary.csv for exact observed type, missing count, missing percentage and cardinality.

Variable	Meaning	Conceptual type	Notebook treatment
encounter_id	Encounter identifier	Integer ID	Excluded from predictors
patient_nbr	Patient identifier	Integer ID	Excluded from predictors; grouping variable
race	Patient race category	Categorical	Missing → Unknown; one-hot encoded
gender	Recorded gender	Categorical	Invalid/Unknown rows removed; retained as feature
age	Age band	Ordinal categorical	Also mapped to age_numeric; banded approximation
weight	Recorded weight category	Categorical	Dropped; 96.86% missing
admission_type_id	Admission type code	Categorical code	Converted to categorical; one-hot encoded
discharge_disposition_id	Discharge disposition code	Categorical code	Categorical; selected codes excluded from rows
admission_source_id	Admission source code	Categorical code	Converted to categorical; one-hot encoded
time_in_hospital	Length of stay in days	Numeric count	Retained; scaled in preprocessing
payer_code	Payer category code	Categorical	Missing → Unknown
medical_specialty	Admitting physician specialty	Categorical	Missing → Unknown
num_lab_procedures	Number of lab procedures	Numeric count	Retained; scaled
num_procedures	Number of procedures	Numeric count	Retained; scaled
num_medications	Number of medications	Numeric count	Retained; scaled
number_outpatient	Prior outpatient visits	Numeric count	Retained; used in total_prior_visits
number_emergency	Prior emergency visits	Numeric count	Retained; used in total_prior_visits
number_inpatient	Prior inpatient visits	Numeric count	Retained; used in total_prior_visits / has_prior_inpatient
diag_1	Primary diagnosis code	Categorical code	Mapped to diag_1_group

diag_2	Secondary diagnosis code	Categorical code	Mapped to diag_2_group
diag_3	Additional diagnosis code	Categorical code	Mapped to diag_3_group
number_diagnoses	Number of diagnoses	Numeric count	Retained; scaled
max_glu_serum	Maximum glucose serum result category	Categorical	Missing → Not tested
A1Cresult	HbA1c result category	Categorical	Missing → Not tested
metformin	Metformin medication change/status	Categorical	Retained; one-hot encoded
repaglinide	Repaglinide medication change/status	Categorical	Retained; one-hot encoded
nateglinide	Nateglinide medication change/status	Categorical	Retained; one-hot encoded
chlorpropamide	Chlorpropamide medication change/status	Categorical	Retained; one-hot encoded
glimepiride	Glimepiride medication change/status	Categorical	Retained; one-hot encoded
acetohexamide	Acetohexamide medication change/status	Categorical	Retained; one-hot encoded
glipizide	Glipizide medication change/status	Categorical	Retained; one-hot encoded
glyburide	Glyburide medication change/status	Categorical	Retained; one-hot encoded
tolbutamide	Tolbutamide medication change/status	Categorical	Retained; one-hot encoded
pioglitazone	Pioglitazone medication change/status	Categorical	Retained; one-hot encoded
rosiglitazone	Rosiglitazone medication change/status	Categorical	Retained; one-hot encoded
acarbose	Acarbose medication change/status	Categorical	Retained; one-hot encoded
miglitol	Miglitol medication change/status	Categorical	Retained; one-hot encoded
troglitazone	Troglitazone medication change/status	Categorical	Retained; one-hot encoded
tolazamide	Tolazamide medication change/status	Categorical	Retained; one-hot encoded
examide	Examide medication indicator/status	Categorical	Retained as recorded; one-hot encoded
citoglipton	Citoglipton medication indicator/status	Categorical	Retained as recorded; one-hot encoded
insulin	Insulin medication change/status	Categorical	Retained; one-hot encoded
glyburide-metformin	Combination medication change/status	Categorical	Retained; one-hot encoded
glipizide-metformin	Combination medication	Categorical	Retained; one-hot encoded

	change/status		
glimepiride-pioglitazone	Combination medication indicator/status	Categorical	Retained as recorded; one-hot encoded
metformin-rosiglitazone	Combination medication indicator/status	Categorical	Retained as recorded; one-hot encoded
metformin-pioglitazone	Combination medication indicator/status	Categorical	Retained as recorded; one-hot encoded
change	Whether diabetes medication changed	Categorical	Retained; one-hot encoded
diabetesMed	Whether any diabetes medication prescribed	Categorical	Retained; one-hot encoded
readmitted	Original readmission outcome (NO, >30, <30)	Categorical target source	Used to derive target; excluded from predictors
early_readmission	Engineered binary target	Binary target	1 = <30; 0 = NO or >30; never predictor

Engineered variables

Variable	Definition	Treatment
total_prior_visits	number_inpatient + number_emergency + number_outpatient	Numeric feature
has_prior_inpatient	1 if number_inpatient > 0, otherwise 0	Binary feature
diag_1_group	ICD-9 chapter grouping of diag_1	Categorical feature
diag_2_group	ICD-9 chapter grouping of diag_2	Categorical feature
diag_3_group	ICD-9 chapter grouping of diag_3	Categorical feature
age_numeric	Numeric mapping derived from age band	Numeric feature; approximate due to grouped age